

RESTING MEMBRANE POTENTIAL (RMP)

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- **Definition :** Electrical potential difference (voltage) across cell membrane (between inside & outside of cell) under resting condition.
- **Also called :—**
 - a. Membrane potential
 - b. Transmembrane potential
 - c. Transmembrane potential difference
 - d. Transmembrane potential gradient
- **RMP of human skeletal muscle :—** -90 mV .

IONIC BASIS OF RMP

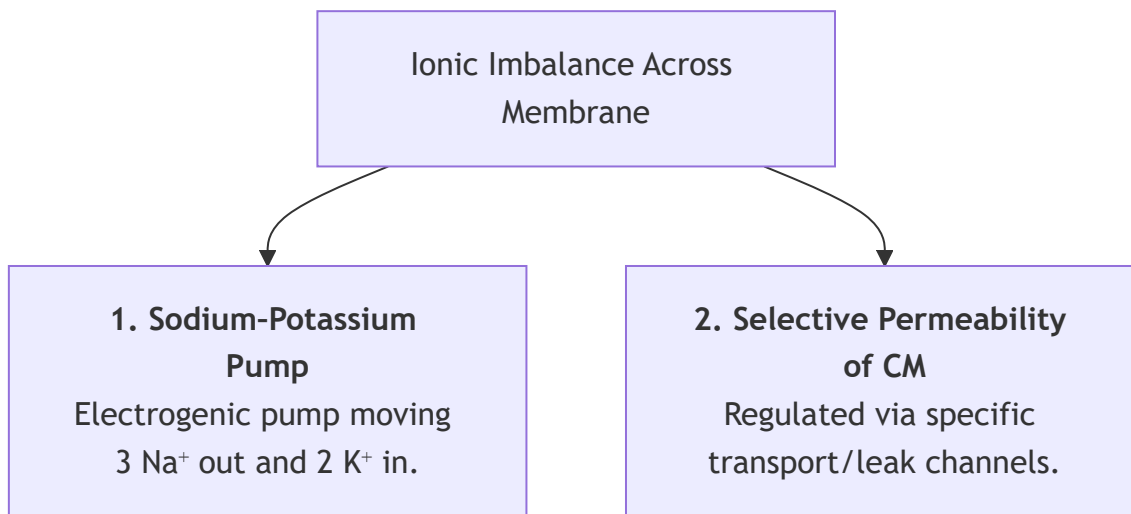
- Development and maintenance of RMP is carried out by movement of ions, which produce ionic imbalance across cell membrane:



Development of more positive outside & more negative inside



Ionic imbalance produced by 2 factors:



1. SODIUM–POTASSIUM PUMP

- Na⁺ & K⁺ ions actively transported in opposite directions across the cell membrane (CM) by means of an **electrogenic pump**:
- Pumps 3 Na⁺ **out** of the cell.
- Pumps 2 K⁺ **into** the cell.
- **Leads to negativity inside & positivity outside of cell.**

2. SELECTIVE PERMEABILITY OF CELL MEMBRANE (CM)

- Permeability of cell membrane depends largely on transport channels (*selective for movement of specific ions*).
- **2 Types of Channels :—**
 - a. Channels for major anions like proteins
 - b. Leak channels

A. Channels for Major Anions Like Proteins

Negatively charged large substances [Proteins, Organic phosphate & Sulfate compounds]



Channels are absent / closed



Substances remain inside cell (Development & maintenance of negativity inside cell)

B. Leak Channels

- **Passive channels** that maintain resting membrane potential by allowing movement of positive ions (Na^+ & K^+) across cell membrane.
- **Distribution of 3 ions (Na^+ , Cl^- , and K^+) :—**
 - Na^+ and Cl^- are more outside.
 - K^+ is more inside.
 - Cl^- ion channels are mostly closed in resting conditions (*retained outside*).

Na^+ is actively transported OUT of cell (against concentration gradient)

K^+ is actively transported INTO cell (against concentration gradient)



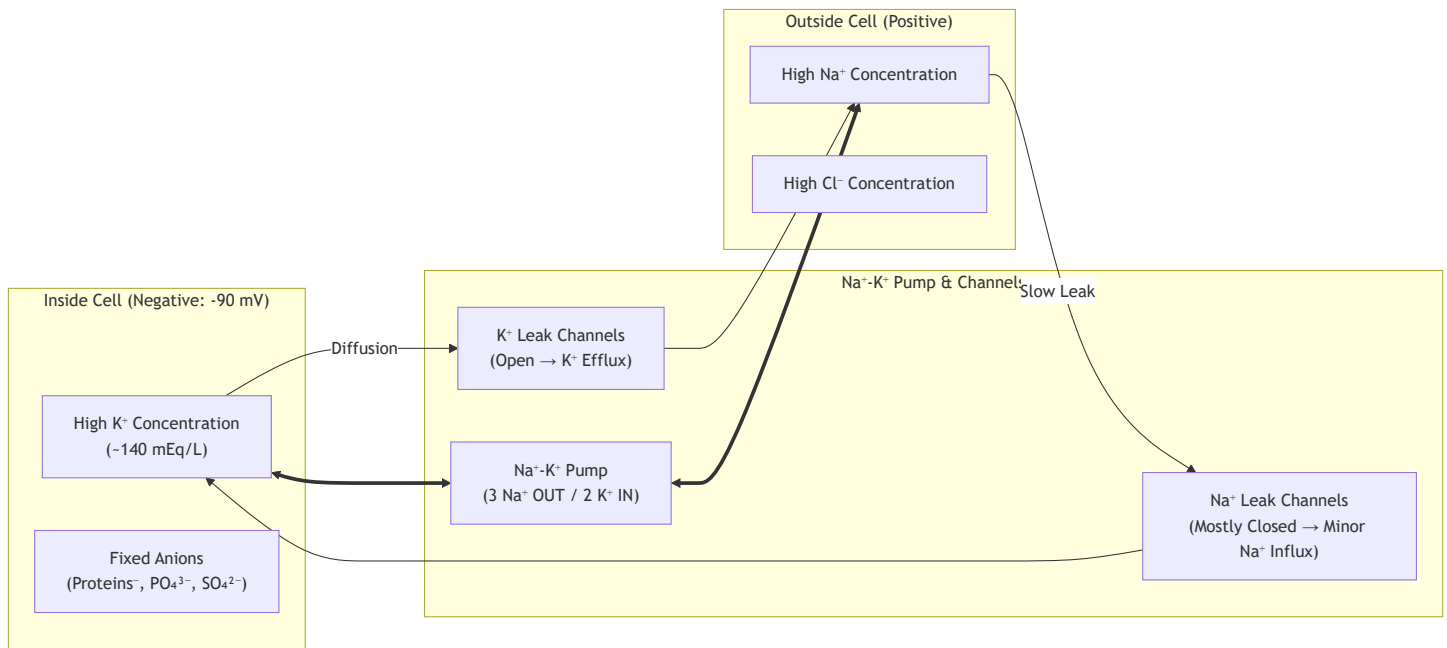
Because of concentration gradient:

Na^+ diffuses back INTO cell through Na^+ leak channels

K^+ diffuses OUT of cell through K^+ leak channels

RESTING STATE PERMEABILITY

- **In resting conditions :—**
 - Almost all K^+ leak channels are opened.
 - Most of Na^+ leak channels are closed.
 - ↓
 - Maintain concentration gradient / equilibrium.



STOPPAGE OF K⁺ EFFLUX

- **After establishment of RMP, efflux of K⁺ ions stops because :—**
- a. Positivity outside of cell repels positive K⁺ ions.
- b. Negativity inside cell attracts positive K⁺ ions.

KEY VALUES & APPLIED ASPECTS

- **Concentration of K⁺ inside cell :—** $\approx 140 \text{ mEq/L}$.
- **Negativity inside fibers :—**
- Inside muscle fiber = -90 mV
- Inside nerve fiber = -70 mV

Hyperpolarization :—

If K⁺ is not present / decreased, negativity inside increases beyond -120 mV .

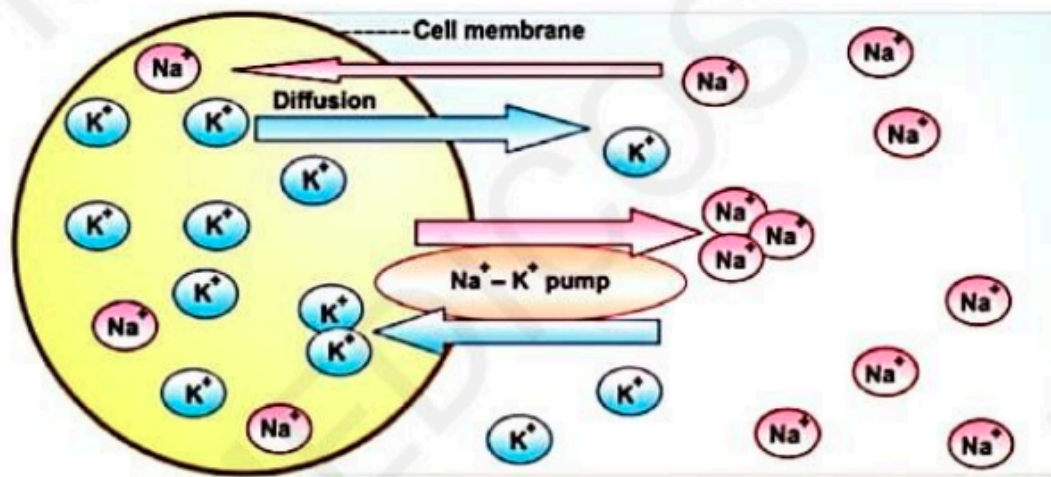


FIGURE 31.1: Development of resting membrane potential by sodium-potassium ($Na^+ - K^+$) pump and diffusion of ions. $Na^+ - K^+$ pump actively pumps three Na^+ outside and two K^+ into the cell. However, the diffusion of K^+ out of the cell is many times greater than the diffusion of Na^+ inside the cell because many of the K^+ leak channels are opened and many of the Na^+ leak channels are closed.